LARS JANSEN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior **AI/Backend Engineer** in the maritime & logistics operations space, building production **agentic** apps with **Python 3.10–3.12** and Azure (**Azure OpenAI**, **AI Foundry**, **Functions**, **Logic Apps**) to automate decisions and cut waste. I deliver **RAG** + **semantic search** with guardrails, tool/function calling, and deterministic business rules, plus reliable APIs in **.NET 6–8** and web UIs in **Angular 14–18**. Strong on MLOps, observability, and incident-driven hardening: I debug real errors, migrate legacy services to modern runtimes, and ship measurable improvements for crews and ops teams across regions, with clear documentation and training.

SKILLS

- **AI / Agentic Systems**: **Azure AI Foundry**, **Azure OpenAI**, prompt patterns (system/few-shot), **tool/function calling**, multi-step workflows, **RAG** (naive→hybrid), embeddings, semantic search, reranking patterns, **guardrails**, deterministic policy checks, eval harnesses, feedback loops, NLP, OCR (Azure AI Vision / Document Intelligence), speech (Azure Speech)
- **Languages**: **Python 3.8–3.12**, **C# 8–12**, SQL, TypeScript 4–5, JavaScript (ES2019–ES2024)
- **Backend & APIs**: **FastAPI 0.9–0.11**, **.NET 6–8**, ASP.NET Core Web APIs, Azure SDKs, REST, OpenAPI, webhooks, background workers, retries/backoff, idempotency, rate limiting, caching, error contracts
- **Cloud & Automation**: Microsoft **Azure**, **Azure Functions**, **Logic Apps**, Azure DevOps Pipelines, Git, Docker 20–26, Kubernetes 1.24–1.29 (concepts/ops), Infrastructure-as-Code patterns (ARM/Bicep/Terraform-style modules)
- **Data**: **Azure SQL**, **SQL Server 2016–2022**, PostgreSQL 12–16, **Azure Cache for Redis (Redis 6–7)**, indexing, query tuning, ETL/ELT, data quality checks
- **Frontend (Plus)**: **Angular 14–18**, RxJS 6–7, component libraries, forms, dashboards, role-based UI, API-driven tables, performance profiling
- **Observability & Quality**: **OpenTelemetry**, Application Insights, structured logging, metrics/SLOs, pytest, xUnit, contract tests, CI quality gates, secure secrets handling, OAuth2/OIDC + **JWT**, RBAC, audit logging

WORK HISTORY

Senior Software Engineer — August 2021 to November 2025
RavenTrack

- I built a production **agentic RAG** service that answered “why” questions on purchasing/consumption by retrieving from Azure SQL + document stores and enforcing deterministic policy steps before responding.
- I integrated **Azure OpenAI** tool/function calling with a rule-engine layer so pricing approvals, substitutions, and compliance checks stayed consistent under edge-case prompts and noisy data.
- I delivered workflow automation using **Azure Logic Apps** to orchestrate invoice ingestion → OCR extraction → validation → exception queues, cutting manual triage while keeping full auditability.
- I fixed a recurring “silent mismatch” bug where OCR field normalization caused wrong SKU mappings, solving it with schema constraints, confidence thresholds, and human-in-the-loop review gates.
- I built **Azure Functions** HTTP APIs for crew-facing and ops-facing actions, adding idempotency keys and replay-safe handlers to prevent duplicate orders during network retries.
- I improved retrieval accuracy by switching from naïve top-k to hybrid ranking (filters + embeddings + rerank rules), reducing hallucinated substitutions and increasing traceable citations.
- I implemented a guardrails layer that blocked tool calls on unsafe inputs and required deterministic confirmations for budget-impacting actions, preventing costly automation mistakes.
- I diagnosed latency spikes caused by chat context bloat and unbounded retrieval, fixing it with chunking limits, caching, and query prefilters that kept responses stable at peak load.
- I migrated a legacy pipeline into a more modern **Python 3.10–3.12** runtime, refactoring brittle parsing into typed models and adding contract tests to prevent regressions.
- I introduced end-to-end monitoring with structured logs, correlation IDs, and SLO dashboards, making failures actionable and speeding incident resolution across time zones.
- I designed a cost-control strategy for LLM calls (prompt budgets, batching, caching, fallback models), preventing runaway spend during high-volume operational periods.
- I partnered with stakeholders across multiple regions to turn ambiguous operational rules into deterministic workflows, reducing back-and-forth and increasing adoption by non-technical teams.
- I created an evaluation harness with golden datasets and regression checks so prompt and retrieval changes shipped safely without breaking business rule compliance.
- I improved data reliability by tightening SQL constraints and adding safe migrations, eliminating inconsistent “delivered vs consumed” reports that confused procurement decisions.
- I mentored engineers on agentic patterns, testing strategy, and secure integration practices, raising the team’s ability to ship reliable automation quickly.

Software Engineer June 2018 to July 2021
Infinite Loop

- I built data-heavy operational dashboards and APIs that turned raw transactional events into actionable KPIs, focusing on correctness, explainability, and performance under real usage.
- I implemented background processing for long-running ingestion tasks, adding retries, backoff, and dead-letter handling to avoid stuck jobs and cascading failures.
- I fixed a critical integration issue where inconsistent JSON contracts broke downstream consumers, solving it with strict validation, versioned schemas, and contract tests.
- I improved SQL performance by redesigning indexes and query plans, eliminating timeouts on peak reporting windows and stabilizing user-facing analytics.
- I delivered caching strategies with **Redis** to reduce repeated reads for hot endpoints, balancing freshness rules with measurable latency improvements.
- I built secure REST endpoints with strong auth patterns and audit logging, ensuring operational actions were traceable and compliant with internal controls.
- I reduced deployment risk by standardizing CI checks and release steps, preventing “works on my machine” bugs and improving rollback reliability.
- I added observability to key workflows through structured logs and metrics, making intermittent errors reproducible and shortening debugging cycles.
- I implemented a robust import pipeline for messy supplier files, using validation + quarantine flows so bad data didn’t poison downstream reporting.
- I collaborated closely with operations to translate edge cases into test cases, raising quality while keeping delivery speed high.
- I continuously refactored legacy modules into clearer boundaries, improving maintainability and onboarding for new contributors.

Software Developer October 2015 to May 2018
Datamart

- I built and maintained business reporting pipelines on **SQL Server**, transforming inconsistent operational inputs into reliable datasets used for daily decision-making.
- I created robust ETL-style jobs with validation and reconciliation checks, preventing silent data drift that previously caused confusing KPI swings.
- I fixed recurring production issues caused by lock contention and long-running queries, solving them with indexing, batching, and safer transaction scopes.
- I implemented export/import tooling that handled malformed CSV/Excel inputs gracefully, adding quarantine flows so operators could correct errors without blocking the pipeline.
- I improved system reliability by adding structured error logging and repeatable runbooks, reducing downtime during incidents and speeding recovery.
- I automated routine operational tasks with scripts and scheduled jobs, reducing manual effort and minimizing human error in repetitive workflows.
- I collaborated with stakeholders to define “single source of truth” rules, aligning reports across teams and eliminating conflicting numbers.
- I improved data access patterns by introducing role-based permissions and controlled views, ensuring sensitive information stayed protected while still usable.
- I contributed to incremental modernization of legacy components by refactoring risky modules into smaller units with tests, lowering change risk over time.

EDUCATION

Bachelor’s degree in Computer Science 2011 to 2015
University of Amsterdam

- Built strong foundations in software engineering, data structures, and systems design, with coursework that supports designing reliable services and data-driven applications.

PROJECTS

Maritime Ops Copilot (Agentic Automation) — Built an **Azure OpenAI** + **Azure AI Foundry** copilot that answers procurement/consumption questions with **RAG**, triggers approved actions via **Logic Apps**, and enforces deterministic rules for budgets, substitutions, and approvals; included OCR-based document intake, audit logs, and regression evaluations to keep outputs stable as policies evolved.